

Myung Chul (Michael) Kim

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Summary

I am a PhD candidate in Mechanical Engineering at Stanford University working on rare event simulation, stochastic dynamics, and scientific machine learning. My research focuses on accelerating rare transitions between metastable states in high-dimensional atomistic systems.

Education

Stanford University, PhD in Mechanical Engineering Sept. 2023 – June 2029
(Expected)

- Research Advisor: Professor Wei Cai
- Research Area: Monte Carlo, Molecular Dynamics, Statistical Mechanics
- Relevant Coursework: Numerical Linear Algebra, Partial Differential Equations, Stochastic Differential Equations, Finite Element Analysis, High Performance Computing

KAIST, BS in Mechanical Engineering and BS in Electrical Engineering Mar. 2019 – Aug. 2023

Georgia Institute of Technology, Exchange Program in Mechanical Engineering Jan. 2022 – Dec. 2022

Research Experience

PhD Candidate, Stanford University – Stanford, CA Sept. 2023 – Present

- Development of theory and computational methods for neural network–driven importance sampling to accelerate Markov chain Monte Carlo and Langevin dynamics simulations of rare events.
- Design of an error-correcting estimator and branching random walk algorithm to recover unbiased transition rates and channel-resolved probabilities.
- Demonstration on atomistic benchmarks, including the Lennard–Jones-7 cluster, resolving the overall transition rate and the relative probabilities among competing reaction channels across temperatures.
- Application of importance sampling guided accelerated Langevin dynamics to protein conformational changes and adatom surface diffusion on crystalline materials.
- Polymer network modeling using coarse-grained molecular dynamics (CGMD) simulation to study mechanical behaviors of polymers under various loading conditions.

Research Assistant, Georgia Institute of Technology – Atlanta, GA Jan. 2022 – Dec. 2022

- Conceptualization, design, and fabrication of soft electronic stretchable pressure sensor for tactile sensing using curved and inter–digitated serpentine structure.
- Development of closed–loop controllable mask fit adjusting mechanism and fabrication of flexible gas sensors for automatic fit-adjustable smart filtering mask.
- Design of flexible PCBs and 3D-printed custom mechanical components for integration with soft electronic devices, optimizing for seamless assembly in wearable electronics applications.

Publications

Accelerated Langevin Dynamics Simulation via Neural Network–Driven Importance Sampling Feb. 2026

Michael M. Kim, Wei Cai

chemRxiv preprint (Submission to Nature Computational Science planned June 2026)

Accelerated Markov Chain Monte Carlo Simulation via Neural Network–Driven Importance Sampling Jan. 2026

Michael M. Kim, Wei Cai

arXiv preprint (Submission to Journal of Chemical Physics planned July 2026)

Smart Filtering Facepiece Respirator with Self-Adaptive Fit and Wireless Humidity Monitoring	Mar. 2025
Kangkyu Kwon, Yoon Jae Lee, Yeongju Jung, Ira Soltis, Yewon Na, Lissette Romero, <u>Myung Chul Kim</u> , Nathan Rodeheaver, Hodam Kim, Chaewon Lee, Seung-Hwan Ko, Jinwoo Lee, Woon-Hong Yeo	
<i>Biomaterials</i> DOI: 10.1016/j.biomaterials.2024.122866	
Stretchable Wearable Wireless Bioelectronics Using All Printed Pressure Sensors and Strain Gauges	Oct. 2024
Nathan Zavanelli, Yoon Jae Lee, <u>Myungchul Kim</u> , Allison Bateman, Matthew Guess, Hyeonseok Kim, Dinesh K. Patel, Woon-Hong Yeo	
<i>Advanced Materials Technologies</i> DOI: 10.1002/admt.202400998	
[Featured as Journal Cover]	
Advances in Electrochemical Sensors for Detecting Analytes in Biofluids	Mar. 2023
Jimin Lee, <u>Myung Chul Kim</u> , Ira Soltis, Sung Hoon Lee, Woon-Hong Yeo	
<i>Advanced Sensor Research</i> DOI: 10.1002/adsr.202200088	
[Featured as Journal Cover]	

Conferences

Includes upcoming events

ChemAI NYC 2026 , Simons Center for Computational Physical Chemistry, New York University - New York, NY	June 2026
• Accelerated Langevin Dynamics Simulation via Neural Network-Driven Importance Sampling	
ICME Research Symposium 2026 , Stanford Institute for Computational & Mathematical Engineering - Stanford, CA	May 2026
• Accelerating Langevin Dynamics Simulation of Rare Events	
AI+Science: Accelerating Discovery , Stanford HAI - Stanford University	May 2026
• Accelerating Rare Event Simulations	
MRS Spring Meeting , Materials Research Society - Honolulu, HI	Apr. 2026
• Neural Network Driven Importance Sampling for Accelerated Kinetic Monte Carlo of Rare Events	
DAMOP 2025 , The 56th Annual Meeting of the APS Division of Atomic, Molecular and Optical Physics - Portland, OR	June 2025
• Accelerating Monte Carlo Simulation of Rare Events by Importance Sampling using Neural Network	
BMES 2022 Annual Meeting , Biomedical Engineering Society - San Antonio, TX	Oct. 2022
• Smart Filtering Facepiece Respirator with Adaptive Fit and Wireless Humidity Monitoring	
Gordon Research Conference , Multifunctional Materials and Structures - Ventura, CA	Sept. 2022
• Stretchable Pressure Sensor using Inter-Digitated Serpentine Structure	

Awards and Honors

James D. Plummer Graduate Fellowship , Stanford University	Sept. 2023 – Aug. 2024
Young Engineers Honors Society , National Academy of Engineering of Korea	Mar. 2020 – Aug. 2023
National Science & Technology Scholarship , Korea Ministry of Science and ICT	Mar. 2019 – Aug. 2023
Global Leadership Award , KAIST	Mar. 2023
Student Design Finalist , Biomedical Engineering Society and Medtronic	Oct. 2022
Open Innovation Challenge Award , Korean Society of Mechanical Engineers	Oct. 2022
President's Undergraduate Research Award , Georgia Institute of Technology	Aug. 2022

Guwon Academic Excellence Scholarship, Guwon Scholarship Foundation

Aug. 2020

KAIST Academic Excellence Scholarship, KAIST

Aug. 2020

Service and Teaching Activities

Teaching Assistant, Stanford University

- ME 123: Computational Engineering (Spring 2026)

Teaching Assistant, Stanford University

- ME 346A: Introduction to Statistical Mechanics (Winter 2025)

Summer Undergraduate Research Fellow Mentor, Stanford University

- Machine Learning Interatomic Potentials for Large Scale Molecular Dynamics (Summer 2025)

Undergraduates Mentored, Stanford University

- Saul Eduardo Perez Herrera (Summer 2025) @ National Autonomous University of Mexico
- Eitan Cohen Arazi (Winter 2024) @ University of Buenos Aires

Patents

A method and apparatus for classification of subtypes of cells with morphological and motility features using hybrid learning

Jan. 2025

Hyunjong Shin, Chanhong Min, Minwoo Kang, Hyuntae Jeong, Taeyoon Kwon, Myung Chul Kim
KR Patent 10-2762542

Skills

Programming Languages: Python, C, MATLAB

Machine Learning: PyTorch, NumPy, SciPy, GPU acceleration

Scientific Computing: Molecular dynamics, Monte Carlo, Finite Element Methods

Softwares / Technologies: LAMMPS, FEniCS, AutoCAD, Solidworks, Altium Designer, PSpice